UMT 312L — Lab F: Hardness testing

Group 1 (Friday) — Mayank Bohat, Chidambar · Lab date 13 November 2026 · Due 27 November 2026
Handout group: **Gp 4** (Al–Cu ageing at 230 °C, 60 min and 30 min)

What this sheet is. Every measurement the handout asks for, in one place. Values that have been measured are printed; values not yet measured are shown as _____. All d_1, d_2 were read on-screen from the optical micrographs taken on 13 Nov; hardness numbers are computed from them, not read off the machine.

Formulae. $HV = 1.8544\,F/d^2$ and $HB = 2F\,/\,\pi D\big(D - \sqrt{D^2 - d^2}\big)$ with ball $D = 2.5$ mm, F in kgf, d in mm.

Still to measure — three blocks, 5 indents each: **(A4)** mild steel, Vickers 100 kgf **(D2)** Al–Cu aged 230 °C 30 min, Vickers **(D3)** Al–Cu aged 230 °C 60 min, Vickers.
Also, blocks A1–A6 have 3 indents where the handout asks for 5.

Part A — effect of material

mild steel, copper, stainless steel; Brinell at one load and Vickers, five indents each

A1 Mild steel — Brinell, 187.5 kgf **[3 of 5]**

#	d_1 (mm)	d_2 (mm)	mean d	HB
1	1.050	1.030	1.0400	210.7
2	1.027	1.038	1.0325	213.9
3	1.044	1.021	1.0325	213.9
4	_____	_____	_____	_____
5	_____	_____	_____	_____
			mean	**212.9**

mean HB = **212.9**, 95 % confidence ±4.6

A2 Copper — Brinell, 187.5 kgf **[3 of 5]**

#	d_1 (mm)	d_2 (mm)	mean d	HB
1	1.474	1.466	1.4700	99.9
2	1.459	1.470	1.4645	100.8
3	1.462	1.480	1.4710	99.8
4	_____	_____	_____	_____
5	_____	_____	_____	_____
			mean	**100.1**

mean HB = **100.1**, 95 % confidence ±1.3

A3 Stainless steel — Brinell, 187.5 kgf **[3 of 5]**

#	d_1 (mm)	d_2 (mm)	mean d	HB
1	0.873	0.869	0.8710	304.8
2	0.879	0.882	0.8805	298.1
3	0.853	0.862	0.8575	314.8
4	_____	_____	_____	_____
5	_____	_____	_____	_____
			mean	**305.9**

mean HB = **305.9**, 95 % confidence ±20.9

A4 Mild steel — Vickers, 100 kgf **[NOT MEASURED]**

#	d_1 (mm)	d_2 (mm)	mean d	HV
1	_____	_____	_____	_____
2	_____	_____	_____	_____
3	_____	_____	_____	_____
4	_____	_____	_____	_____
5	_____	_____	_____	_____
			mean	_____

A5 Copper — Vickers, 100 kgf **[3 of 5]**

#	d_1 (mm)	d_2 (mm)	mean d	HV
1	1.377	1.328	1.3525	101.4
2	1.352	1.349	1.3505	101.7
3	1.386	1.337	1.3615	100.0
4	_____	_____	_____	_____
5	_____	_____	_____	_____
			mean	**101.0**

mean HV = **101.0**, 95 % confidence ±2.2

A6 Stainless steel — Vickers, 100 kgf **[3 of 5]**

#	d_1 (mm)	d_2 (mm)	mean d	HV
1	0.799	0.771	0.7850	300.9
2	0.776	0.766	0.7710	312.0
3	0.791	0.770	0.7805	304.4
4	_____	_____	_____	_____
5	_____	_____	_____	_____
			mean	**305.8**

mean HV = **305.8**, 95 % confidence ±14.0

Part C — effect of load

mild steel, Brinell, three indents at each of five loads

C1 Mild steel — Brinell, 21.2 kgf **[complete]**

#	d_1 (mm)	d_2 (mm)	mean d	HB
1	0.415	0.442	0.4285	145.9
2	0.444	0.443	0.4435	136.1
3	0.448	0.447	0.4475	133.7
			mean	**138.6**

mean HB = **138.6**, 95 % confidence ±16.1

C2 Mild steel — Brinell, 60 kgf **[complete]**

#	d_1 (mm)	d_2 (mm)	mean d	HB
1	0.601	0.608	0.6045	206.0
2	0.588	0.606	0.5970	211.2
3	0.600	0.604	0.6020	207.7
			mean	**208.3**

mean HB = **208.3**, 95 % confidence ±6.7

C3 Mild steel — Brinell, 100 kgf **[complete]**

#	d_1 (mm)	d_2 (mm)	mean d	HB
1	0.754	0.760	0.7570	217.0
2	0.797	0.768	0.7825	202.7
3	0.729	0.730	0.7295	234.0
			mean	**217.9**

mean HB = **217.9**, 95 % confidence ±39.0

C4 Mild steel — Brinell, 150 kgf **[complete]**

#	d_1 (mm)	d_2 (mm)	mean d	HB
1	0.939	0.929	0.9340	211.0
2	0.931	0.946	0.9385	208.9
3	0.935	0.933	0.9340	211.0
			mean	**210.3**

mean HB = **210.3**, 95 % confidence ±3.0

C5 Mild steel — Brinell, 187.5 kgf **[complete]**

#	d_1 (mm)	d_2 (mm)	mean d	HB
1	1.039	1.038	1.0385	211.4
2	1.035	1.031	1.0330	213.7
3	1.042	1.035	1.0385	211.4
			mean	**212.1**

mean HB = **212.1**, 95 % confidence ±3.4

Part D — effect of heat treatment

Al–Cu, Vickers 100 kgf, five indents per condition

D1 Al–Cu as-solutionized (550 °C 1 h, WQ) — Vickers, 100 kgf **[complete]**

#	d_1 (mm)	d_2 (mm)	mean d	HV
1	1.176	1.236	1.2060	127.5
2	1.234	1.187	1.2105	126.6
3	1.245	1.178	1.2115	126.3
4	1.249	1.202	1.2255	123.5
5	1.247	1.189	1.2180	125.0
			mean	**125.9**

mean HV = **125.9**, 95 % confidence ±1.5

D2 Al–Cu aged 230 °C, 30 min — Vickers, 100 kgf **[NOT MEASURED]**

#	d_1 (mm)	d_2 (mm)	mean d	HV
1				
2				
3				
4				
5				
			mean	

D3 Al–Cu aged 230 °C, 60 min — Vickers, 100 kgf **[NOT MEASURED]**

#	d_1 (mm)	d_2 (mm)	mean d	HV
1				
2				
3				
4				
5				
			mean	

Summary of what is measured so far

Material	Condition	HB (187.5 kgf)	HV (100 kgf)	note
Copper	as received	100.1	101.0	HB and HV agree to 1 %
Mild steel	as received	212.9	———	Vickers not yet done
Stainless steel	as received	305.9	305.8	HB and HV agree to 0.03 %
Al–Cu	as-solutionized	—	125.9	±1.5 at 95 % confidence
Al–Cu	aged 230 °C 30 min	—	———	
Al–Cu	aged 230 °C 60 min	—	———	

Effect of load on mild steel (Brinell, 2.5 mm ball).

Load (kgf)	21.2	60	100	150	187.5
mean d (mm)	0.4398	0.6012	0.7563	0.9355	1.0367
HB	**138.6**	208.3	217.9	210.3	212.1

Above 60 kgf the Brinell number is constant at 212 ± 6. The 21.2 kgf point reads 35 % low — the indentation size effect: too small a load and the indent samples only the work-hardened surface layer rather than the bulk.

One point to confirm with the instructor. Blocks A6 and A3 (3 Vickers and 3 Brinell indents) are assumed here to be the *stainless steel*, because they agree at 305.8 HV and 305.9 HB. If those indents were actually on the mild steel, then the missing block A4 is the stainless steel instead. Mild steel measures 212.9 HB, which would correspond to about 223 HV, not 306 — which is why they have been assigned to the stainless.

Source: 36 optical micrographs, each with d_1 and d_2 measured on screen, 500 μm scale bar. Raw readings archived as `readings.json`.